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A new high-pressure cell for equilibrium measurements of systems with fluid and solid phases

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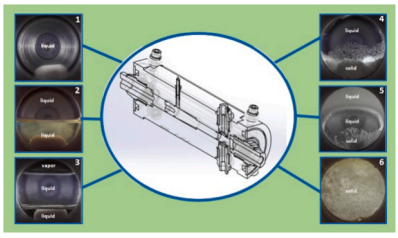
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HIGHLIGHTS

- New PVT cell for high-pressure phase equilibrium measurements.
- Solid-liquid, solid-liquid-liquid, liquid-liquid, and liquid-liquid-vapor phase transitions.
- Heavy alkanes /CO_2 phase behavior and phase diagram at high pressures.

GRAPHICAL ABSTRACT

A high-pressure cell for phase equilibrium measurements



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ABSTRACT

A new high-pressure equilibrium cell to obtain data for systems containing fluids and solids is presented. By using the static synthetic method, an experimental unit was commissioned to obtain phase transition data at pressures up to 1000 bar, and temperatures ranging from 253 to 473 K. Measured phase transition data include new data for liquid-vapor, liquid-liquid, liquid-liquid-vapor, solid-liquid, and solid-liquid-liquid of CO_2 + n-hexadecane, CO_2 + n-eicosane, and CO_2 + n-tetracosane binary systems. Results are compared to those from literature and show that the new experimental cell setup presents a good performance and allows to explore regions where traditionally, accurate experimental data are scarce.

1. Introduction

Knowledge of high-pressure phase equilibrium is a crucial tool for understanding many industrial processes, including the design and optimization of chemical processes, separation process, oil reservoirs management, advanced oil recovery, capture and storage of carbon dioxide, transport, and storage of natural gas, studies of geological processes. Supercritical, ionic liquids, hydrate formation, wax, and asphaltene precipitation [1–3]. Although obtaining accurate experimental data is time-consuming and requires expensive structures, direct measurement of equilibrium data remains an essential source for equipment design and accurate process modeling [2].

Experimental methods used for investigating the phase behavior at high pressure can be divided into two major classes of methods: Analytical and synthetic methods, depending on how the phase compositions are determined and whether the mixture was prepared with

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Table 1
Chemical used in this work.

Chemical Name	CAS Number	Source	Mole % Purity
Methanol	67-56-1	Sigma Aldrich	99.8
n-Hexadecane	544-76-3	Sigma Aldrich	99.0
n-Eicosane	112-95-8	Sigma Aldrich	99.0
n-Tetracosane	646-31-1	Sigma Aldrich	99.0
Carbon dioxide	124-38-9	Air liquid	99.9

known compositions. The experimental methods can be described according to the classification proposed by Dohrn et al. [1].

A commonly used method for phase equilibria measurements at high pressure is the static-synthetic method, this is characterized by the previous preparation of the mixture to know the global composition. The temperature and pressure are then changed until the raising of an incipient phase is observed visually by an appropriate device like a digital microscope. Given the mixture with known global composition, the phase envelope can be traced, including boundaries of the solid, liquid, and vapor regions. The static-synthetic visual method is widely used in the literature for determining the phase envelopes of multicomponent mixtures [2–10], as it does not require a sampling process. Thus, controlling the temperature and pressure of the system is straightforward, and pressure fluctuations are mitigated. This method can be used to investigate any phase transition. Generally, for liquid-liquid, liquid-liquid-vapor, and vapor-liquid, the transition pressure is obtained by setting the temperature and slowly decreasing the pressure until the rising of an incipient phase. This method can be used for phase boundaries determination, including liquid-vapor, liquid--liquid-vapor, liquid-solid, and phase transition in complex systems like hydrates, ionic-liquid and polymeric solutions. For the solid, since phase transition is more sensitive to the temperature, the pressure is fixed, and the temperature is slowly reduced until the nucleation and subsequent precipitation. It is important to notice that the static synthetic visual method does not measure the phase composition at equilibrium. For a given mixture, pressure and temperature conditions of the phase transition are measured for a known global composition.

A known limitation of the visual static method is when the refractive index of the phases in equilibrium are close and visually, the interface is indistinguishable. In this case the non-visual static method like the near infrared spectroscopy has been reported recently in literature [11]. However, the non-visual methods require a further treatment of measured data to obtain the phase envelop and are limited to vapor-liquid transition. In this work, we present the design of an equilibrium cell which use the visual statistical method. The novelties of the cell still on its relatively low cost, an innovative piston design for pressure intensification and a jacket designed for rigorous temperature control necessary for solid phase transition measurements. Using this

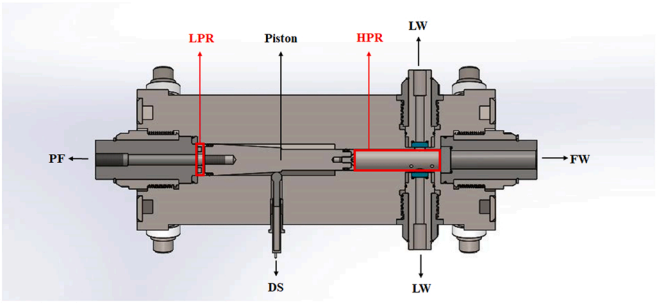


Fig. 2. Internal Scheme of the HPF parts. LPR – Low-Pressure Region; HPR – High-Pressure Region (diameter of 22 mm); DS – Displacement Sensor; LW – Lateral Sapphire Window; FW – Front Sapphire Window; PF – Pneumatic Fluid Inlet.

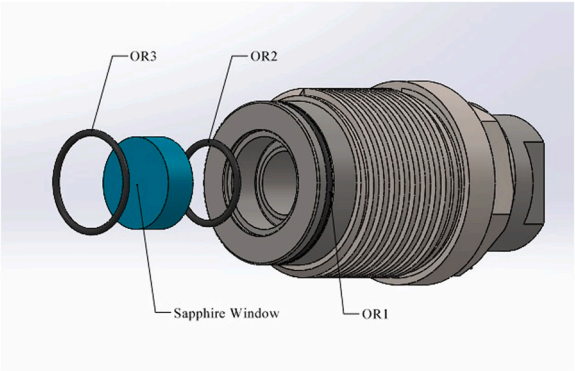


Fig. 3. Side and frontal support for the windows sapphire (OR3 – Oring 3, OR1 – Oring 1, OR2 – Oring 2).

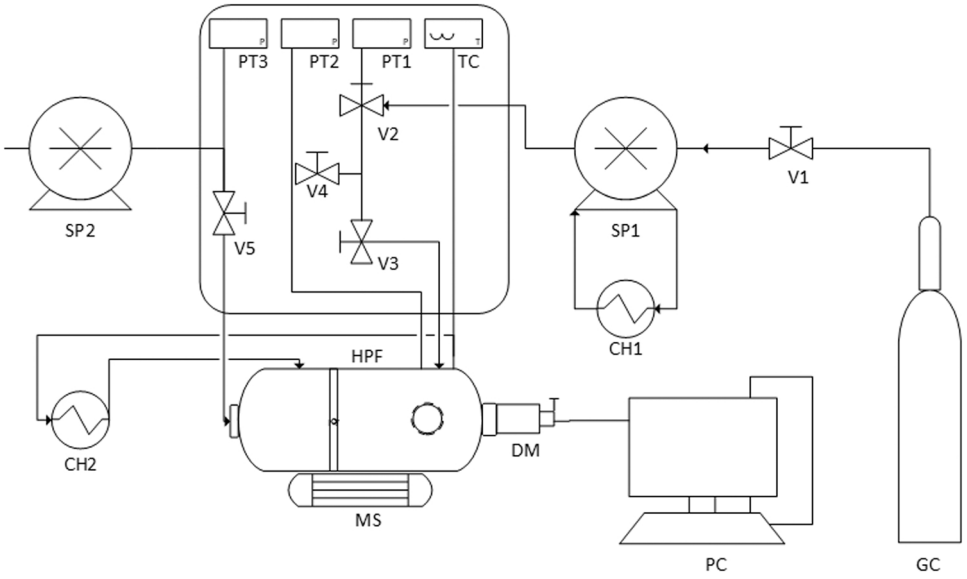


Fig. 1. Experimental Scheme for the static-synthetic method. SP1 and SP2 – High-Pressure Pumps. HPF – Equilibrium Cell. V1, V2, V3, V4 and V5 – Valves. CH1 and CH2 – Thermostatic Baths. DM – Digital Microscope. CG – Gas Cylinder. CP – Computer. MS – Magnetic Stirrer. TC – Temperature Control. PT – Pressure Transducer.

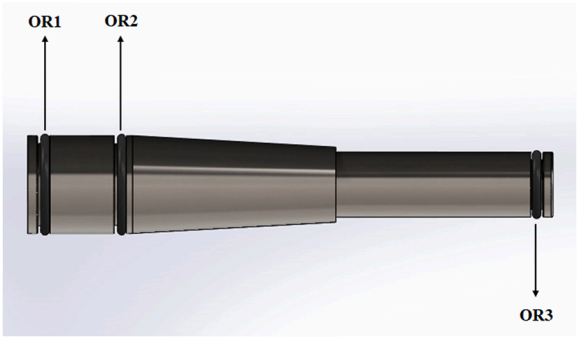


Fig. 4. Movable piston design. The OR holds for O-rings.

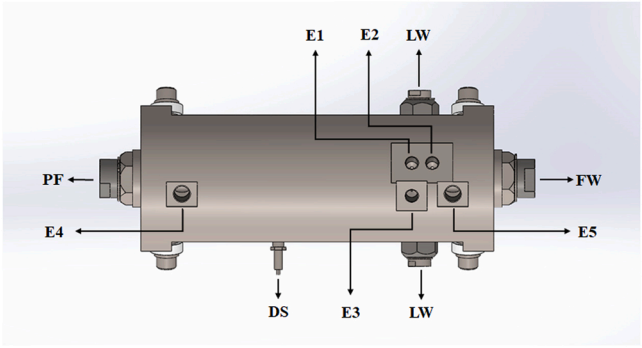


Fig. 5. HPF overall view. PF – Pneumatic Fluid Inlet; E1 – Thermocouple Inlet; E2 – Feeding Inlet; E3 – Pressure Transducer Inlet; E4 – Cooling Inlet; E5 – Cooling Outlet.

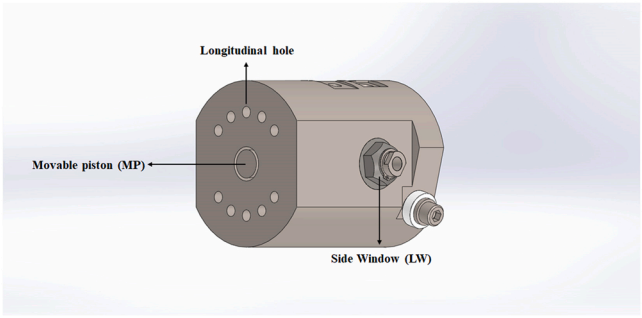


Fig. 6. Cross-section of the equilibrium cell for visualization of longitudinal holes.

cell, we obtained new data of phase transition for liquid-vapor, liquid-liquid, liquid-liquid-vapor, solid-liquid, and solid-liquid-liquid of CO_2 + n-hexadecane, CO_2 + n-eicosane, and CO_2 + n-tetracosane binary systems. In addition to providing experimental data, the systems studied are essential for the oil industry, mainly to flow assurance. Because these hydrocarbons present molar mass like an average molar mass of oil, they can be used as model systems in flow assurance research.

2. Objective

The aim of this work is to present a new high-pressure equilibrium cell that uses the visual static synthetic method suitable to obtain phase transition envelops of complex phase systems. Furthermore, new phase transition data for binary systems containing CO_2 + n-hexadecane, CO_2 + n-eicosane, and CO_2 + n-tetracosane are presented. Measured phase transition data include liquid-vapor (LV), liquid-liquid (LL), liquid-liquid-vapor (LLV), solid-liquid-liquid (SLL), solid-liquid (SL), and solid-liquid-vapor (SLV) equilibria.

Table 2
Liquid-vapor transition data for the system CO_2 + methanol, where x_1 is the carbon dioxide mole fraction.

T/K	u^T/K	x_1	P/bar	u^P/bar
303.0	0.1	0.2384	35.2	1.2
		0.4247	53.7	1.2
		0.5211	57.2	1.2
		0.7182	61.6	1.2
		0.867	63.5	1.2
313.0	0.1	0.2384	41.5	1.2
		0.4247	64.4	1.2
		0.5211	68.9	1.2
		0.7182	75.6	1.2
		0.867	77.8	1.2
323.0	0.1	0.2384	48.3	1.2
		0.4247	73.7	1.2
		0.5211	81.3	1.2
		0.7182	92.3	1.2
		0.867	92.8	1.2
333.0	0.1	0.2384	54.8	1.2
		0.4247	85.0	1.2
		0.5211	95.7	1.2
		0.7182	110.5	1.2
		0.867	108.0	1.2

u^T and u^P are the uncertainties for temperature and pressure and respectively. $u^{x_1} = 0.0003$.
The referred uncertainty is an expanded combined uncertainty, for ($\alpha = 0.05$, $k = 2$),

$$U = k\sqrt{u^2_{(measure)} + u^2_{(Temperaute\ \ or\ \ Pressure\ \ transducer\ \ resolution)}}$$

Where the k is the coverage factor and α is the lower and upper bonds of the confidence interval. $U(T) = 0.9$ K, for the transition from liquid, liquid-liquid, or liquid-liquid-vapor to solid and $U(P) = 1.2$ bar for liquid-liquid, liquid-vapor or liquid-liquid-vapor transitions. The measured uncertainty ($u_{measured}$) is the mean error associated to the triplicate of each experiment.

3. Materials and methods

3.1. Materials

Table 1 shows the chemicals and their mole fraction purity used in this work. All chemicals were used without further purification.

3.2. Experimental method

We used the static-synthetic method described in Section 1. The main feature of this method is to synthesize a mixture with known global composition into an equilibrium cell. The temperature and pressure conditions are then adjusted until the raising of an incipient phase.

Fig. 1 shows the scheme of the experimental apparatus used to obtain the equilibrium data. The experimental unit consists of a variable volume cell (HPF), two high-pressure syringe pumps, two thermostatic baths, a magnetic stirrer, a digital microscope, a gas cylinder, an analytical balance, and a control panel with valves and connections.

3.3. The pressure equilibrium cell

HPF – Equilibrium Cell (Figs. 2–6) was designed and built by the ATOMS group (Applied Thermodynamics and Molecular Simulation) and consisted of a two concentric stainless-steel cylinder (316 L) with a total weight of 40 kg. The cell has a maximum and a minimum working volume equal to 50 mL and 15 mL, respectively. The cell is similar to a shell tube heat exchanger. The internal tube has two regions (see Fig. 2): the Low-Pressure Region (LPR) and the High-Pressure Region (HPR). In the LPR, the internal diameter is 32 mm and a length of 137 mm. This region is connected to the pneumatic fluid from SP2 and to the movable

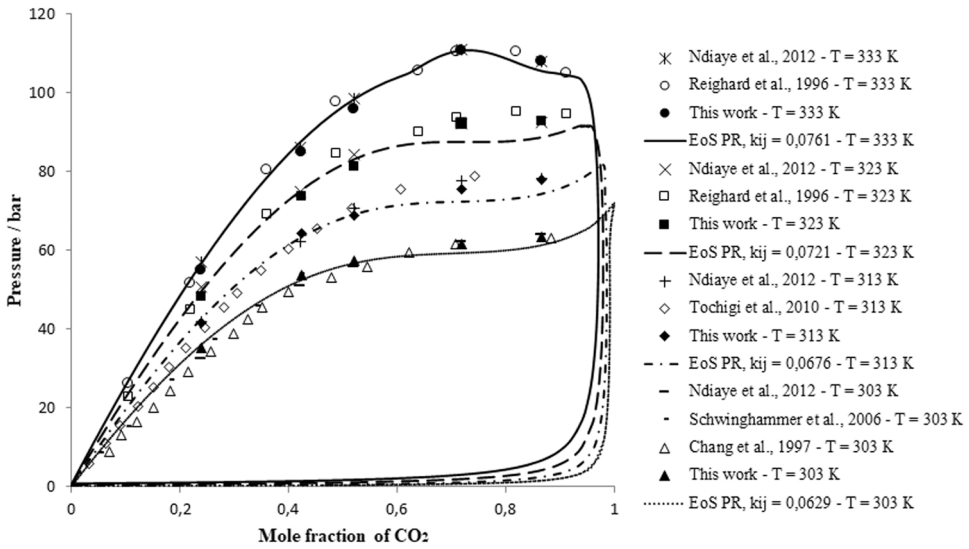


Fig. 7. P-*x* diagram for the system CO_2 + methanol at 303, 323 and 333 K.

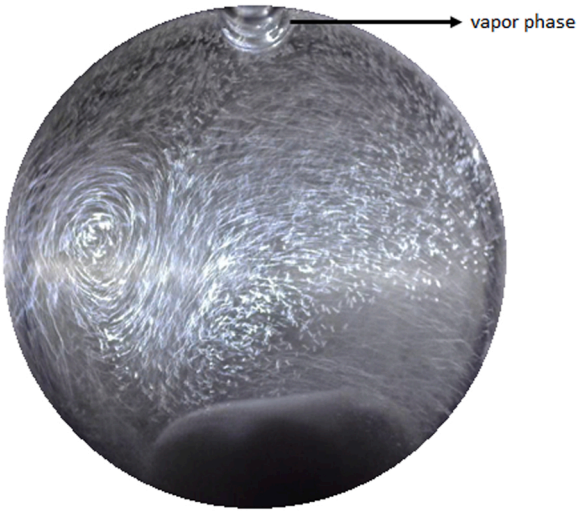


Fig. 8. Image of the liquid-vapor phase transition to the CO_2 + methanol system. The system is not in equilibrium due to the migration of bubbles to the top, forming a vapor phase.

piston. The High-Pressure Region (HPR) diameter is 22 mm, and a length of 111 mm. This region contains the samples and can reach pressures up to 1000 bar, and temperatures ranging from 253 to 473 K. The external tube presents ten longitudinal holes where refrigerant from a chiller can circulate to control the sample temperature precisely.

The HPF cell has a frontal sapphire window (FW) coupled to a digital optical microscopy device to detect the phase transitions. Additionally, the cell has two symmetrical side sapphire windows (SW) – one for illuminating the sample chamber and another for eventual coupling of a non-visual phase transition detection device.

Fig. 3 shows the design of the device containing the side and frontal sapphire windows. Three O-rings (OR) are used: The first O-ring (OR1) allows a static sealing since there is no relative movement between the cell surface and the device. The second and third O-rings (OR2 and OR3) are squeeze sealing. They are allocated respectively behind and in front of the sapphire windows. The sealing consists of the sample pressing the sapphire windows against the OR2, O-ring allocated in the cavity behind the sapphire windows. In this configuration, the higher the sample pressure, the better the sealing.

One of the equilibrium cell innovations is the movable piston (MP) shown in Fig. 4. The MP is set to move in two regions (LPR and HPR) with different areas. This setting allows pressure in the sample chamber three times greater than the pressure applied through the syringe pump behind the piston. Furthermore, the piston has three o-rings (OR), as shown in Fig. 4, for dynamic sealing – two in the LPR (OR1 and OR2) and one in the HPR (OR3).

The use of O-rings for sealing at high pressures must be accompanied by a preliminary study of the temperature and pressure conditions and the components presented in the mixture to determine the most compatible O-ring for a given mixture. Here, mixtures with high carbon dioxide content are used, and pressures up to 503.4 bar were measured with temperature varying from 280.75 to 373.45 K. A compatible and low-cost sealing material is the buna O-ring (nitrile) which showed good performance.

For the phase transitions visualization, a digital microscope (DM) is used. The sample is monitored in real-time, and the incipient phase can be detected instantly with a magnitude of up to 140 times the actual size.

The equilibrium cell has three upper inputs (Fig. 5). Input E1 is connected to a thermocouple. Input 2 (E2) is connected to valve V3 and is used for feeding gas components into the cell. The pressure transducer TP2 is connected to Input 3 (E3) and directly used to obtain the sample pressure. The cell has two additional inputs (E4 and E5) connected to the jacket. Longitudinal holes along the jacket allow a refrigerant fluid to circulate, and a rigorous temperature control range from 253 to 473 K is achieved.

Figs. 5 and 6 show the external view and cross-section of the equilibrium cell, respectively.

In addition to the equilibrium cell, the experimental unit for phase transition data measurements includes the following equipment:

SP1 – High-Pressure Pump (TELEDYNE ISCO Syringe Pump Model 260D) has a cylinder with a capacity of up to 266 mL, the volumetric flow that varies between 0.01 and 107 mL/min, and a pressure of up to 571 bar. This pump is used to inject precise amounts of solvents (CO_2, CH_4, and others) into the equilibrium cell. The pump has a jacket connected to a chiller (CH1) for the pump reservoir temperature control.

SP2 – This pump is the same model as SP1 and is used to connect to the piston. The pump is used to control the sample pressure. The chamber of the pump is filled with deionized water as a pneumatic fluid.

CH1 and CH2 – Thermostatic Bath (Julabo CORIO CD – 200F) has a volume of 4 L and is connected to the equilibrium cell to control the sample temperature. The chiller can operate with temperatures ranging from 253 to 423 K, with a precision of 0.1 K.

Table 3
Experimental phase transition data for the CO_2 + n-hexadecane where x_1 is the global carbon dioxide mole fraction.

T/K	u^T/K	P/bar	u^P/bar	Transition
$x_1 = 0.9231$				
297.8	0.2	388.2	1.2	L→LL
298.0	0.2	321.7	1.2	L→LL
299.5	0.2	255.9	1.2	L→LL
302.7	0.2	189.5	1.2	L→LL
307.8	0.1	166.1	1.2	L→LL
313.1	0.1	159.9	1.2	L→LL
323.1	0.2	161.0	1.2	L→LL
333.1	0.3	171.3	1.2	L→LL
343.1	0.2	185.6	1.2	L→LL
353.1	0.3	199.8	1.2	L→LL
363.1	0.2	213.5	1.2	L→LL
284.2	0.2	38.9	1.2	LL→LLV
288.0	0.1	42.9	1.2	LL→LLV
293.1	0.2	49.2	1.2	LL→LLV
298.1	0.1	55.5	1.2	LL→LLV
303.1	0.2	62.5	1.2	LL→LLV
308.1	0.2	70.7	1.2	LL→LLV
280.8	0.2	42.5	1.2	LL→SLL
282.4	0.3	122.0	1.2	LL→SLL
283.9	0.2	199.2	1.2	LL→SLL
285.1	0.3	255.2	1.2	LL→SLL
286.4	0.4	321.5	1.2	LL→SLL
288.4	0.2	386.6	1.2	LL→SLL
$x_1 = 0.9537$				
294.5	0.2	321.6	1.2	L→LL
294.6	0.2	388.1	1.2	L→LL
295.3	0.2	255.8	1.2	L→LL
296.1	0.1	210.2	1.2	L→LL
298.4	0.1	189.0	1.2	L→LL
303.3	0.3	163.6	1.2	L→LL
308.3	0.2	148.4	1.2	L→LL
313.2	0.2	144.3	1.2	L→LL
323.1	0.1	159.5	1.2	L→LL
333.1	0.1	171.3	1.2	L→LL
343.1	0.3	185.6	1.2	L→LL
353.1	0.2	199.8	1.2	L→LL
363.1	0.1	213.5	1.2	L→LL
285.3	0.1	40.1	1.2	LL→LLV
288.1	0.1	44.1	1.2	LL→LLV
293.1	0.2	51.0	1.2	LL→LLV
298.1	0.1	59.0	1.2	LL→LLV
303.1	0.1	66.0	1.2	LL→LLV
308.1	0.2	74.5	1.2	LL→LLV
313.4	0.9	81.5	1.2	LL→LLV
280.7	0.9	50.7	1.2	LL→SLL
282.1	0.9	121.7	1.2	LL→SLL
283.6	0.9	189.2	1.2	LL→SLL
285.0	0.9	255.7	1.2	LL→SLL
286.3	0.9	321.5	1.2	LL→SLL
287.5	0.9	386.7	1.2	LL→SLL
$x_1 = 0.9789$				
284.0	0.1	53.4	1.2	L→LL
288.2	0.1	61.3	1.2	L→LL
291.6	0.2	67.2	1.2	L→LL
293.2	0.2	70.6	1.2	L→LL
298.1	0.1	82.9	1.2	L→LL
303.1	0.2	95.5	1.2	L→LL
308.1	0.2	105.2	1.2	L→LL
313.1	0.2	116.5	1.2	L→LL
323.1	0.1	133.3	1.2	L→LL
333.1	0.3	152.2	1.2	L→LL
343.2	0.2	169.4	1.2	L→LL
353.1	0.3	185.2	1.2	L→LL
363.2	0.3	198.9	1.2	L→LL
283.8	0.2	37.9	1.2	LL→LLV
288.1	0.3	43.2	1.2	LL→LLV
293.1	0.3	49.1	1.2	LL→LLV
298.1	0.1	56.1	1.2	LL→LLV
303.1	0.2	63.2	1.2	LL→LLV
308.0	0.2	70.1	1.2	LL→LLV
281.4	0.9	55.1	1.2	L→SL
282.4	0.9	121.3	1.2	L→SL
283.4	0.9	190.2	1.2	L→SL
284.6	0.9	255.1	1.2	L→SL
285.9	0.9	321.7	1.2	L→SL
287.2	0.9	387.0	1.2	L→SL

u^T and u^P are the uncertainties for temperature and pressure, respectively.
$u^{x_1} = 0.0003$.

MS – Magnetic Stirrer (IKA C_MAG MS 10) is used to agitate the sample to keep the homogeneity.

DM – Digital Microscope (Dino-Lite AM4515ZTL) allows the visualization of the phase transitions inside the equilibrium cell. With a resolution of 1.3 megapixels and magnification up 140 times the actual size, this device provides high-quality images and allows the detection of any incipient phases in the sample mixture. The DM is connected to a computer for video recording. It is important to emphasize that the HPF equilibrium cell is a full view cell and the digital covers the entire surface of the sapphire window. Thus, any event like nucleation bubbles is detected and recorded.

CG – A gas cylinder is used to store gases used in the experiment. The cylinder is connected to the high-pressure pump (SP1).

CP – Control Panel is a structure that houses the pressure and temperature indicator and also valves (V2, V3, V4, V5). V2 is used to isolate the high-pressure pump and avoid the sample displacement into the syringe pump chamber in case of failure of V3. V3 is a micrometric valve used to supply the compressed CO_2 or methane into the equilibrium cell. V4 is a relief valve used for safety in case of leakage. V5 is connected to the piston and, when opened, allows the piston displacement to control the sample pressure. PT1 is a pressure indicator used to indicate the pressure during the feeding. The Pressure Transducer PT2 indicates the sample pressure, while PT3 indicates the pneumatic fluid pressure. All pressure transducers are Swagelok model S with pressure range from 0 to 100 MPa and accuracy of 0.02 MPa. The piston is designed so that the pressure indicated by PT2 is three times the pressure showed by PT3. T1 is a temperature indicator and is connected to a thermocouple immersed in the sample. V1 check valve used to feed the SP1 chamber with gas from the cylinder (CG).

The Analytical Balance (Shimadzu. Model AUX220) has a maximum capacity of 1220 g with an accuracy of 0.1 mg. The balance is used to weigh the substances that are liquid or solid at room temperature and pressure.

4. Experimental procedure

The experimental procedure for obtaining the equilibrium data is based on the static-synthetic method shown in Fig. 2 and described in previous works [4–6] for more details.

The process starts by loading the syringe pump SP1 chamber with gas. Because at room temperature (298 K), merely opening the gas cylinder valve is not sufficient to move enough solvent into the SP1 chamber, the following procedure is used to fulfill the syringe pump. The cylinder is opened for about forty minutes, and the valve V2 is kept closed. The temperature of the SP1 jacket is reduced to 283 K. This arrangement allows natural flow from the cylinder at room temperature to the pump chamber SP1 syringe kept at 283 K through the thermostatic bath (CH1). A typical experiment begins by injecting precise amounts of each substance into the cell. For liquid charging, first, the cell is flushed with dry air, and the vacuum pump is turned on to remove any traces of dirt inside the cell. The liquid is then introduced into the cell, using a sampling cylinder and a high precision balance. The exact mass of liquid or solid component introduced is determined with a high precision balance with an accuracy of 0.1 mg. The CO_2 is charged using a syringe pump (SP1) stabilized at 150 bar and 290 K. At these conditions, carbon dioxide is liquid with a known density (0.92074 g/mL).

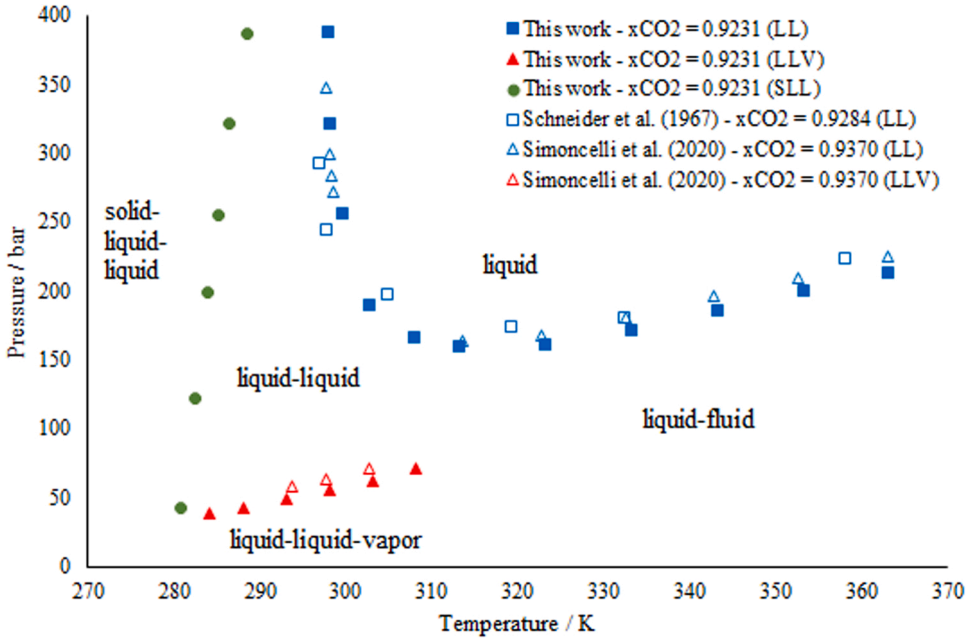


Fig. 9. P-T diagram for the system CO_2 + n-hexadecane. For fixed CO_2 mole fraction at 0.9231.

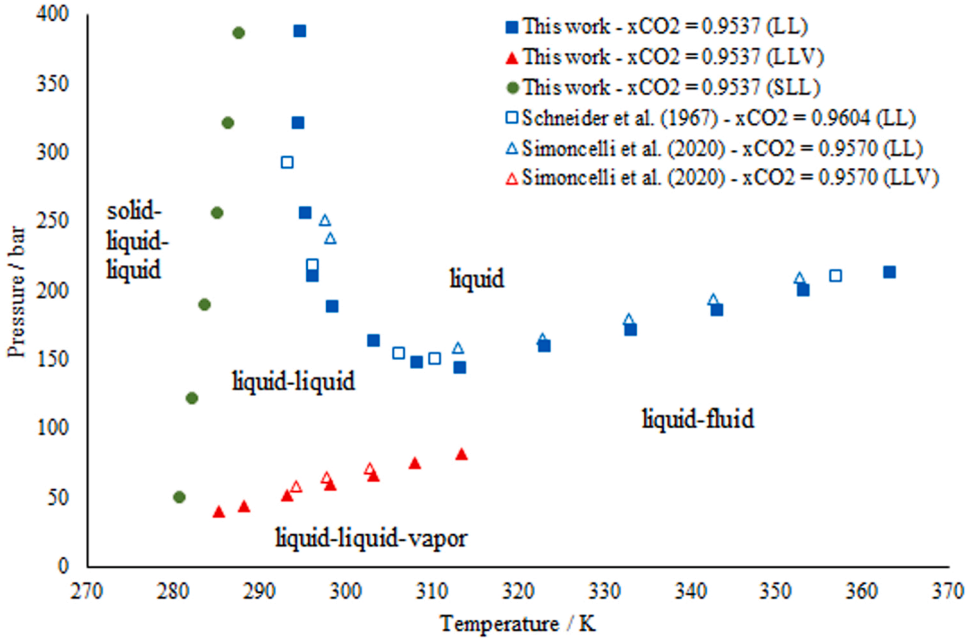


Fig. 10. P-T diagram for the system CO_2 + n-hexadecane. The mole fraction of CO_2 is fixed at 0.9537.

The standard uncertainties u in mass introduced were estimated for liquid and solid, with $u = 0.0004$ g and $u = 0.02$ g for CO_2, respectively. The cell is then closed and connected to the process line, keeping valves V1, V3 and V4 closed and V2 open. The line is pressurized using the SP1 and is stabilized at 150 bar and 290 K when carbon dioxide is used in the sample mixture. At these conditions, the density is 0.92074 g/mL. With the density, the mass of carbon dioxide for a known composition can be converted to the volume of CO_2. System stabilization (zero pump flow) requires about thirty minutes and must be done carefully because any trace of flow can lead to systematic errors in determining the volume of CO_2 to be injected. Once the system is stabilized, a given volume of gas is injected into the cell through the micrometric valve V3 and the syringe pump controller SP1. Once this is done, the SP2 is connected to the piston. Finally, the procedures for obtaining equilibrium data are initiated. In this work, we use a pressure transducer from Swagelok (model S) calibrated by the company. We also used a type J thermocouple, calibrated using an ASTM thermometer.

4.1. Experimental procedures for liquid-vapor, liquid-liquid, and liquid-liquid-vapor equilibria

To obtain liquid-vapor (LV), liquid-liquid (LL), and liquid-liquid-vapor (LLV) transition data, a sequence of procedures is initiated to obtain a single-phase system. The temperature is adjusted to a given value circulation bath CH2. Valve V5 is opened, while valves V1, V2, V3, and V4 are kept closed. SP2 is used to displace the piston, that is, to

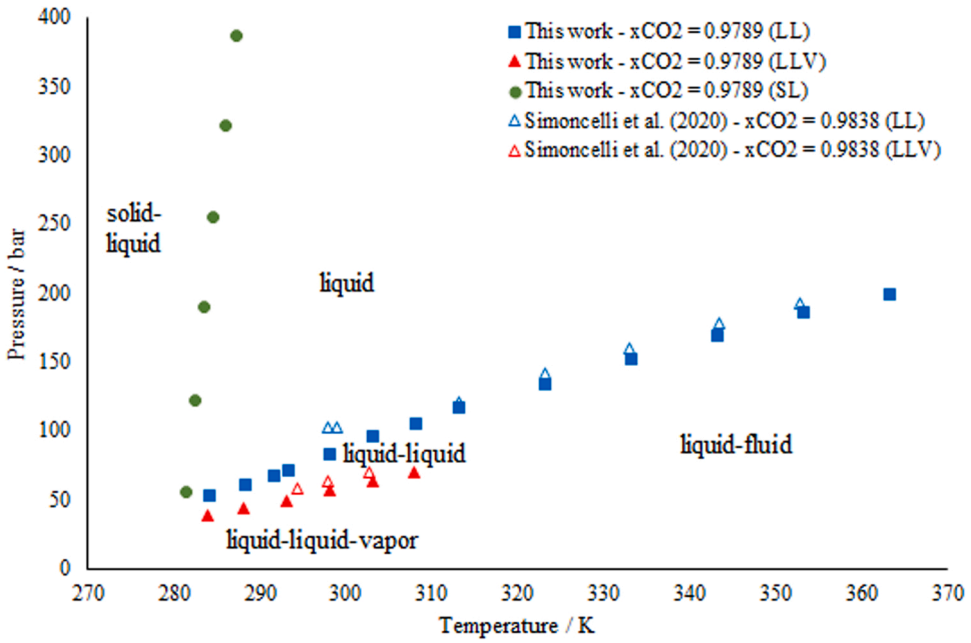


Fig. 11. P-T diagram for the system CO_2 + n-hexadecane. The mole fraction CO_2 is fixed at 0.9789.

pressurize and depressurize the sample. With the mixture continuously under agitation, the pressure is gradually increased to reach the condition of a single-phase system. With keeping the temperature fixed, the SP2 controller is used to gradually reduce the pressure at a rate of 0.5 bar/min until the digital microscope detects a new incipient phase formation. The temperature and pressure conditions at which the transitions occur, with the known overall composition, constitute the phase envelope's point. This procedure is repeated at least three times for each temperature and fixed global composition. Then, the temperature is set to a new value, and the entire measurement process is repeated. For liquid-liquid-vapor measurement (LLV), the liquid-liquid transition is first measured, and the pressure is reduced up to the raising of the vapor phase.

4.2. Experimental procedures for solid-liquid equilibria

For transitions in which the solid phase is present because the solid-liquid transition is more sensitive to the temperature, the pressure is fixed in a region where the mixture is homogeneous or in liquid-liquid or liquid-vapor equilibrium. Under continuous agitation, the temperature is reduced slowly, employing the thermostatic bath (CH2) connected to the equilibrium cell. Once the solid-liquid transition point is obtained, maintaining the constant pressure, the temperature is increased until the solid phase's disappearance, and after twenty minutes under agitation, the previous procedure is repeated. This procedure is performed at least three times for each given pressure. The cooling rate for the formation of crystals was 0.8 °C/min.

4.3. Validation of the equilibrium cell apparatus

Table 2 presents the experimental data obtained at a temperature ranging from 303 to 333 K, with phase transition pressures varying from 35 to 111 bar, with combined uncertainty of 0.9 K and 1.2 bar, for temperature and pressure, respectively.

The results obtained were compared to those from literature and the Peng-Robinson equation of state calculations. Fig. 7 presents the experimental data from this work, from literature, and thermodynamic modeling [4,7–10]. As can be observed, the data obtained here from the proposed apparatus agree with those from the literature and are consistent with the EoS calculations.

Fig. 8 shows an image of the liquid-vapor phase equilibrium of the CO_2 + methanol system. One can see the formation of a vapor phase and the migration of bubbles to the top. This figure shows how the digital microscope can enhance the quality of phase transition visualization inside the equilibrium cell.

To check the equilibrium cell performance for transition type liquid-liquid, liquid-liquid-vapor, and solid-liquid and solid-liquid-vapor, we use the system CO_2 + n-hexadecane and compare the results to those from the literature. Data from literature were taken from Schneider et al. [12] and Simoncelli et al. [13] who used the static synthetic visual method.

5. Results

5.1. CO_2 + n-hexadecane

For the CO_2 + n-hexadecane system, phase transition experimental data were obtained, with temperatures ranging from 280.7 to 363.2 K and pressure up and 388.2 bar. The experimental data include liquid-liquid (LL), liquid-liquid-vapor (LLV), solid-liquid (SL), solid-liquid-liquid (SLL) equilibria. Table 3 shows the phase transition pressure for the fixed temperature and CO_2 mole fraction. At these compositions, transitions observed are liquid to liquid-liquid (L → LL), liquid-liquid to liquid-liquid-vapor (LL → LLV), liquid to solid-liquid (L → SL), and liquid-liquid to solid-liquid-liquid (LL → SLL).

Figs. 9 and 10 show the pressure versus temperature diagram for the CO_2 + n-hexadecane system with carbon dioxide mole fractions fixed at 0.9231 and 0.9537. Data for liquid transition are compared to those from Schneider et al. [12]. According to the phase diagram classification by Konynenburg and Scott (1980) [14], one can observe a type III behavior of both isotherms. The liquid-liquid transition curve (LL) changes continuously to the fluid-liquid transition curve above the carbon dioxide critical temperature, with a minimum. The liquid-liquid-vapor (LLV) curve starts at the low-temperature region and ends at the upper critical endpoint, at temperatures closes to the carbon dioxide critical temperature. The solid-liquid-liquid curve is observed at low temperatures, and for a fixed temperature, the pressure at which the solid phase transition appears is independent of the carbon dioxide concentration.

The major advantage of the equilibrium cell presented in this work,

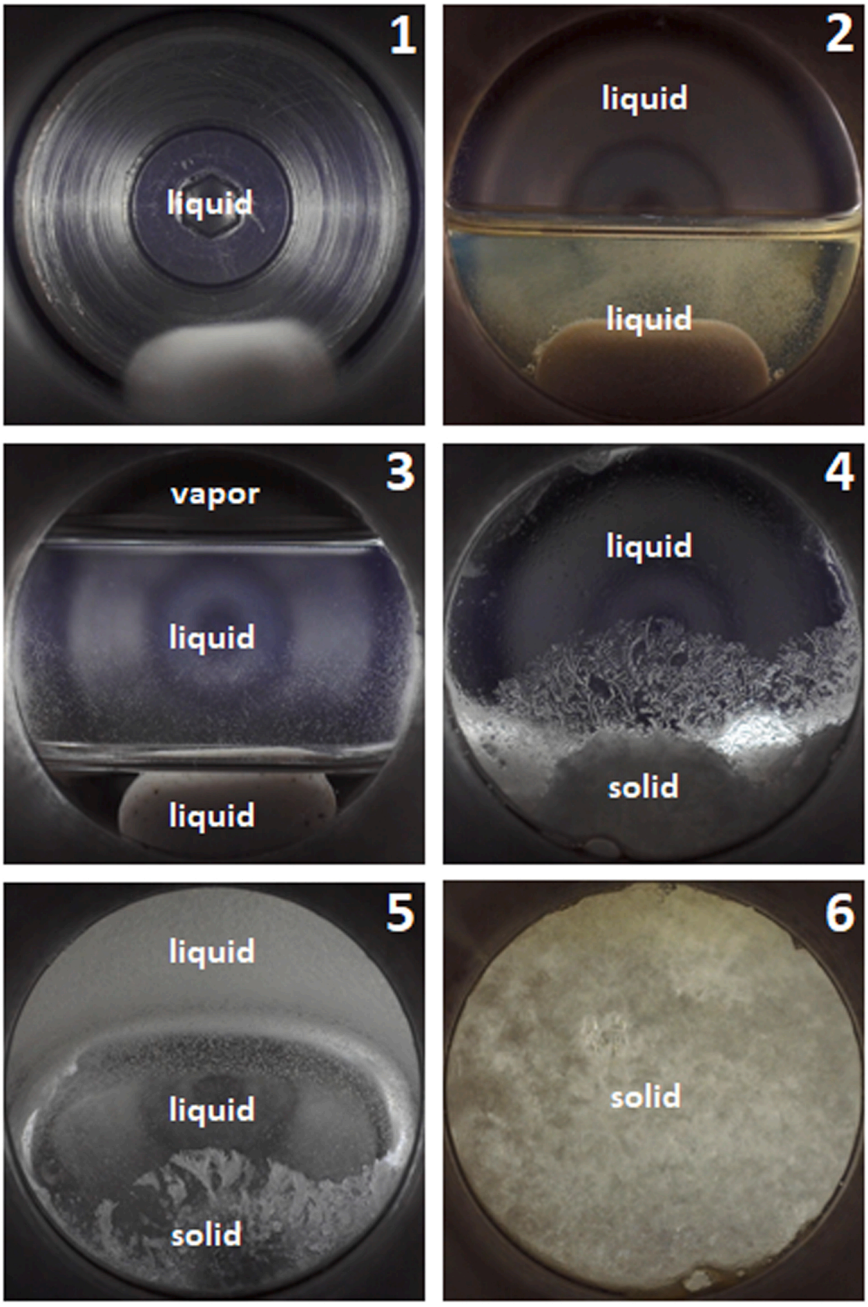


Fig. 12. Images of the phase transitions to the CO_2 + n-hexadecane system. We show different phases, including vapor, liquid, and solid. The conditions are: **1** – x_1 = 0.9231, 250 bar and 323.1 K; **2** – x_1 = 0.9231, 120 bar and 333.1 K; **3** – x_1 = 0.9537, 45 bar and 303.1 K; **4** – x_1 = 0.9789, 321.7 bar and 281.2 K; **5** – x_1 = 0.9231, 255.2 bar and 280.1 K; **6** – x_1 = 0.9789, 321.7 bar and 277.3 K.

is the time saving during experimental run. For a defined global composition, the survey of the phase envelope in the P-T plane is done in a relatively short time when compared to other devices which use the static synthetic visual method. An additional advantage still on the solid phase detection. The design of the cell allows to fix the pressure at a given value and then vary the temperature of the cell using a thermal fluid that circulates through the longitudinal holes of the cell. This scheme allows for strict temperature control and enables the detection of solid- liquid, solid-liquid, and the solid-liquid-vapor transitions.

By increasing the carbon dioxide mole fraction to 0.9789, different behavior is observed for the same system (Fig. 11). In contrast to the previous curve, the pressure of liquid-liquid transition increases when increasing temperature without a minimum in the P-T diagram. The solid-liquid, liquid-fluid, and liquid-liquid-vapor curves converge to a solid-liquid-vapor point.

Fig. 12 shows images of the phase transitions obtained by the HPF equilibrium cell proposed here for the mixture CO_2 + n-hexadecane under different temperatures, pressures, and compositions. Image 1 presents the system as entirely homogeneous. Only the liquid phase is visualized. The conditions for the appearance of this single phase are at high pressures. In the experiments studied for the CO_2 + n-hexadecane system, it was possible to observe one homogeneous phase under pressure conditions above 53.4 bar for 0.9789 CO_2 mole fraction and above 144.3 bar for 0.9231 and 0.9537 CO_2 mole fractions. Image 2 displays the system with two liquid phases. The upper phase, rich in CO_2, has a smaller density than the lower phase, rich in n-hexadecane. Image 3 presents three distinct phases. A denser liquid phase at the bottom. A less dense liquid phase at the central part. Besides a vapor phase, located at

Table 4
Experimental phase transition data for the CO_2 + n-eicosane where x_1 is the carbon dioxide mole fraction.

T/K	u^T/K	P/bar	u^P/bar	Transition
$x_1 = 0.9625$				
318.6	0.3	479.9	1.2	L→LL
318.6	0.1	438.7	1.2	L→LL
319.6	0.2	400.0	1.2	L→LL
322.1	0.3	334.4	1.2	L→LL
333.1	0.2	286.0	1.2	L→LL
338.1	0.2	281.0	1.2	L→LL
343.1	0.3	291.0	1.2	L→LL
348.0	0.2	293.0	1.2	L→LL
353.2	0.1	291.8	1.2	L→LL
301.1	0.2	63.1	1.2	LL→LLV
302.2	0.2	64.0	1.2	LL→LLV
303.4	0.3	65.7	1.2	LL→LLV
303.9	0.2	66.8	1.2	LL→LLV
305.3	0.1	68.3	1.2	LL→LLV
306.6	0.2	71.2	1.2	LL→LLV
300.1	0.9	60.9	1.2	LL→SLL
301.4	0.9	160.0	1.2	LL→SLL
303.0	0.9	239.0	1.2	LL→SLL
304.5	0.9	319.4	1.2	LL→SLL

u^T and u^P are the uncertainties for temperature and pressure, respectively.
$u^{x_1} = 0.0003$.

the top. Image 4 shows a system with two distinct phase, a solid at the bottom and a liquid at the top. For systems with heavy hydrocarbons, the temperature is the main property for the appearance of paraffinic crystals. Image 5 shows a three-phase system, two liquids and one solid. Systems in these conditions are observed by reducing the temperature from a region with two distinct liquid phases, maintaining the pressure constant. In this image, it is possible to observe the curvature between the liquid phases, different from the usual. This phenomenon is related to the proximity with the so-called isodensity, i.e., two phases with similar densities. Image 6 presents a rich solid phase system. This condition is observed when the temperature is below the solid-liquid transition temperature.

5.2. CO_2 + n-eicosane

Experimental data for the CO_2 + n-eicosane system is reported in Table 4. The observed behavior is similar to that of the CO_2 + n-hexadecane system with higher pressures for all transition types, as shown in Fig. 13. There is a shift in pressure and temperature by increasing the hydrocarbon chain, especially for the transition from the liquid to liquid-liquid.

5.3. CO_2 + n-tetracosane

Table 5 presents experimental data for the CO_2 + n-tetracosane system. Under the conditions studied, it was possible to see the transitions of the liquid-liquid, solid-liquid-liquid, and solid-liquid-vapor type for a mole fraction of CO_2 equal to 0.9685.

Fig. 14 shows the CO_2 + n-tetracosane system with liquid-liquid transitions at high pressures and temperatures. When compared with the CO_2 + n-hexadecane and CO_2 + n-eicosane systems, it is possible to

Table 5
Experimental phase transition data for the CO_2 + n-tetracosane where x_1 is the carbon dioxide mole fraction.

T/K	u^T/K	P/bar	u^P/bar	Transition
$x_1 = 0.9685$				
349.5	0.2	503.4	1.2	L→LL
348.8	0.2	482.6	1.2	L→LL
350.1	0.3	469.6	1.2	L→LL
352.4	0.3	445.9	1.2	L→LL
353.5	0.2	437.9	1.2	L→LL
358.7	0.2	427.3	1.2	L→LL
363.5	0.2	413.0	1.2	L→LL
368.5	0.3	403.5	1.2	L→LL
373.4	0.3	396.0	1.2	L→LL
313.1	0.9	85.0	1.2	LL→SLL
313.9	0.9	121.9	1.2	LL→SLL
314.9	0.9	201.6	1.2	LL→SLL
317.6	0.9	307.8	1.2	LL→SLL
319.4	0.9	413.1	1.2	LL→SLL
321.8	0.9	517.9	1.2	LL→SLL

u^T and u^P are the uncertainties for temperature and pressure, respectively.
$u^{x_1} = 0.0003$.

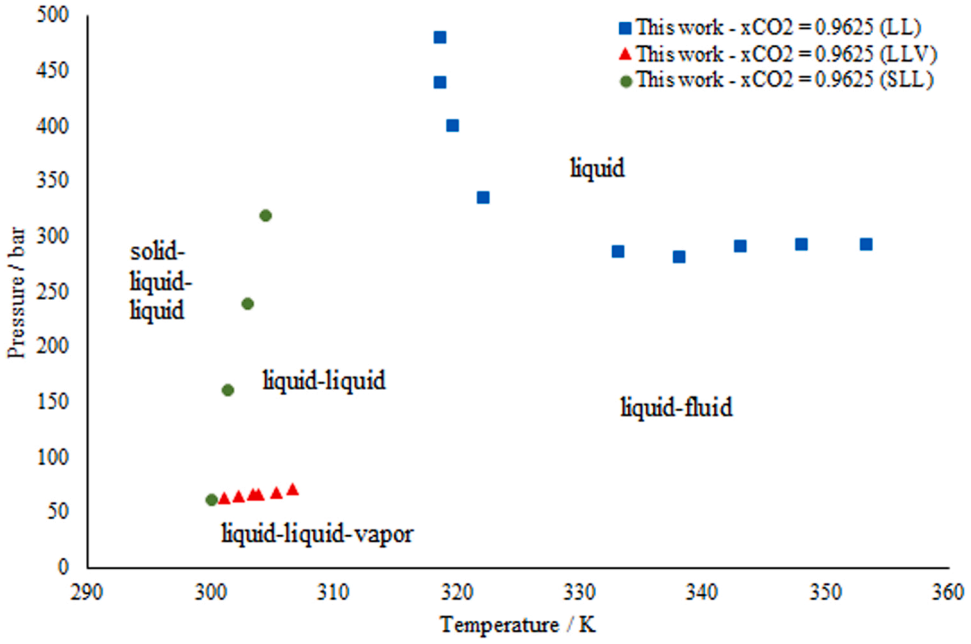


Fig. 13. P-T diagram for the system CO_2 + n-eicosane with CO_2 mole fraction fixed at 0.9625.

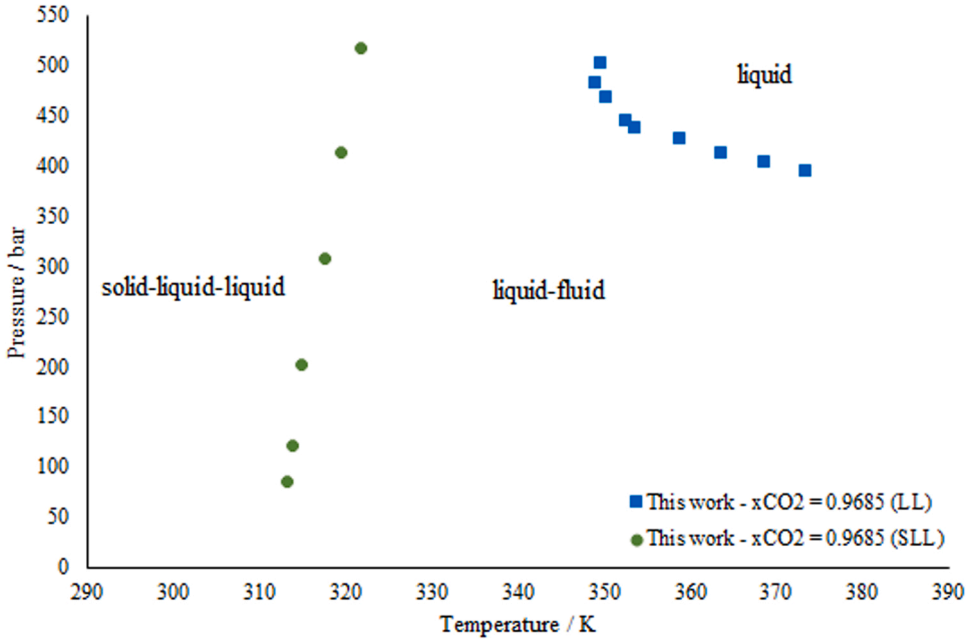


Fig. 14. P-T diagram for the system CO_2 + n-tetracosane with CO_2 mole fraction fixed at 0.9685.

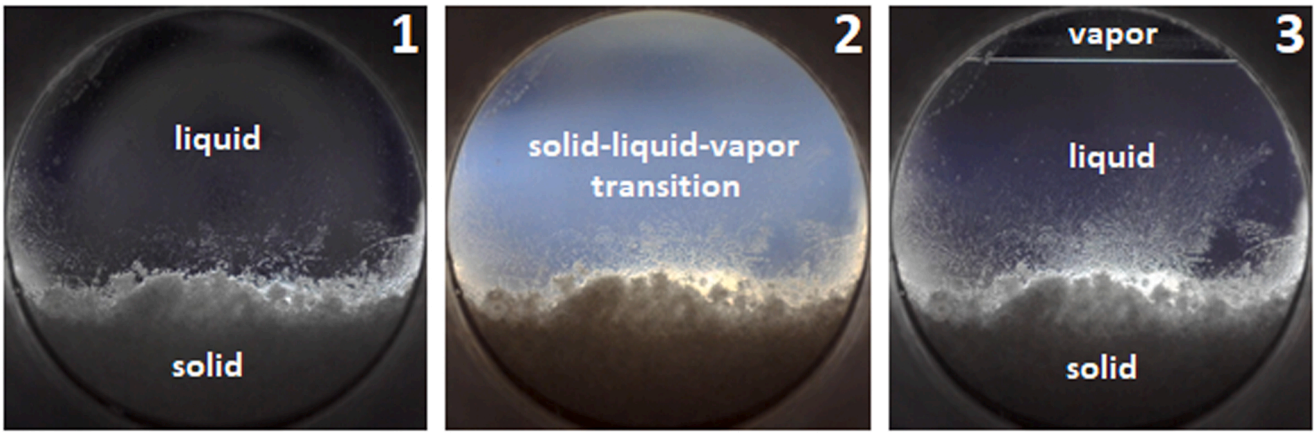


Fig. 15. Images of the solid-liquid and solid-liquid-vapor transitions for the CO_2 + n-tetracosane mixture.

observe the influence of the hydrocarbon chain size on the phase transitions, shifted to higher pressures and temperatures. Here, due to the limitations of experimental conditions, it was impossible to see the transitions below 280.7 K. However, for the specific case of CO_2 + n-tetracosane mixture, it was possible to visualize the solid-liquid-vapor transition.

Fig. 15 shows the solid-liquid and the solid-liquid-vapor, including the point at incipient vapor phase appearance, for the CO_2 + n-tetracosane mixture. Image 1 shows the system with two phases, solid and liquid. The system is at 90 bar and 305.9 K. The temperature is kept constant, and the pressure is reduced until it reaches image 2. At this point occurs the solid-liquid-vapor transition at a pressure of 70 bar, where the turbidity is observed. Image 3 shows three well-defined phases, observed due to the continuous pressure reduction, keeping the temperature constant.

6. Conclusions

A new high-pressure equilibrium cell was designed, and the static-synthetic method was used to obtain phase-transition data including liquid-vapor (LV), liquid-liquid (LL), solid-liquid (SL), liquid-liquid-vapor (LLV), solid-liquid-liquid (SLL), and solid-liquid-vapor (SLV), for the CO_2 + n-hexadecane, CO_2 + n-eicosane, and CO_2 + n-tetracosane binary mixtures. Results show that the high-pressure equilibrium cell can reproduce data from literature with high accuracy. Additionally, we provided new equilibrium data for the CO_2 + n-hexadecane, CO_2 + n-eicosane, and CO_2 + n-tetracosane systems at very high CO_2 concentrations. The investigated systems exhibited a type III behavior, according to the Konynenburg and Scott classification [14].

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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